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PALEOMAGNETIC AND PETROGRAPHIC INVESTIGATION OF THE TAUM SAUK LIMESTONE, SOUTHEAST MISSOURI

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Abstract. Paleomagnetic results from 17 sites in one quarry of the Cambrian (Dresbachian Stage) Taum Sauk Limestone (member Bonneterre Formation) reveal two components of magnetization. On the basis of thermal demagnetization, coercivity spectrum analysis, and petrographic evidence, both components reside in hematite. Sites can be divided into three groups based on behavior during thermal demagnetization and petrographic criteria. The first group consists of massive pink limestones and contains a component with declination 270°, inclination -11°, corresponding to a Cambrian pole at 175.6°E, 3.6°S. The second group of sampling sites consists of specimens with numerous burrows filled with hematite intergrown with calcite. The sites are dominated by the second component of magnetization which has declination 147° and inclination 10° and corresponds to a reversed pole position at 133.1°E, 37.3°N. A significant fraction of this component, which is interpreted to be Carboniferous in age, resides in hematite intergrown with calcite in the burrows. Petrographic relationships indicate that the calcite and hematite were acquired before dolomite, which places a maximum age on dolomitization. The third group consists of massive pink limestones with variable amounts of burrowing and contains both directions of magnetization in roughly equal proportions, with closely overlapping blocking temperature spectra. The presence of two stable directions of magnetization from one location with neither being dominant or even present throughout the quarry indicates the significant effect that diagenetic and sedimentologic processes may have on the magnetization of sedimentary units.

Introduction

Remagnetization of sedimentary rocks by diagenetic processes is important in the history of many units. Remagnetization can considerably complicate the magnetic history of such units, particularly when the natural remanent magnetization (NRM) is the vector sum of two or more components. One example of this may be where a detrital remanent magnetization (DRM) exists in conjunction with a chemical remanent magnetization (CRM) which was acquired after deposition. Although the objective of many paleomagnetic investigations is the isolation of a detrital remanence associated with the unit's origin to calculate depositional paleopoles, isolation of secondary CRM can also provide valuable information on the diagenetic history of sedimentary units.

This study was undertaken to investigate the Taum Sauk Limestone which contains abundant diagenetic hematite intergrown with calcite [Frank, 1981]. The specific objective was to date the time of hematite precipitation. As discussed below, this study demonstrates the complexity of magnetization in sedimentary rocks, especially where remagnetization has occurred. The study also demonstrates that dating of diagenetic events is possible by paleomagnetic means under some circumstances. In addition, results were obtained that pertain to the Cambrian section of the apparent polar wander path (APWP) of North America.

Geologic Setting and Previous Work

All sampling was done within one quarry of the Taum Sauk Limestone in southeast Missouri (Figure 1). The Upper Cambrian Taum Sauk is a member of the Bonneterre Formation (Dresbachian Stage) and is limited in areal extent to the St. Francois Mountains in southeast Missouri. First described by Brightman [1938], the Taum Sauk has irregular distribution, thickness, and stratigraphic position within the Bonneterre. This is attributed to its accumulation in basins of an archipelagic complex formed by the hills of Precambrian igneous rocks [Brightman, 1938]. The Bonneterre lies conformably on the Cambrian Lamotte Sandstone and is the host of lead mineralization in the Southeast Missouri Mineral District.

Howe [1968] recognized what had previously been called the Taum Sauk as one facies with alternating shallow marine carbonate muds and stromatolitic layers (planar stromatolite and burrowed carbonate mud facies) (Figure 2). Larsen [1977] interpreted the environment of deposition for this facies as having a low-energy level and forming in a high intertidal to supratidal zone. The low-energy environment resulted from baffling of northwesterly winds by the Precambrian hills [Howe, 1968] and protection by a reef which partially encircled the archipelago [Gerdemann and Myers, 1972]. The planar stromatolite and burrowed carbonate mud facies flank the Precambrian igneous mountains and extend across formation boundaries [Howe, 1968]. Owing to the lack of exposed section or datum markers, proper age assignment of exposed sections of the Taum Sauk is difficult. However, the quarry where sampling was done has been correlated to the Bonneterre Formation by Howe [1968] and is certainly no older than Dresbachian.

Most previous work on the Bonneterre has been conducted from a diagenetic and/or depositional environment perspective or has been done in an effort to understand better the Mississippi Valley type (MVT) lead mineralization in the

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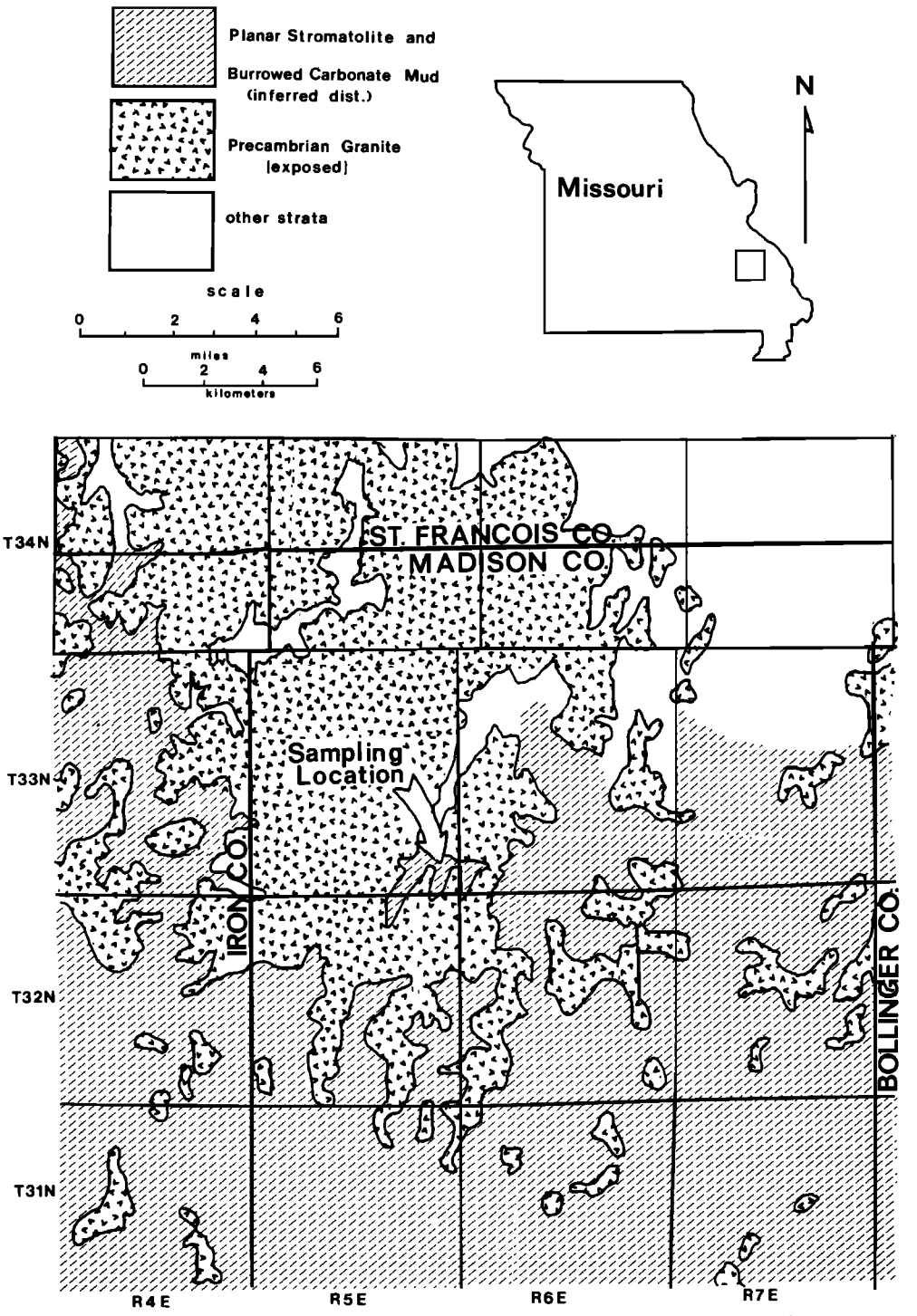


Fig. 1. Location map in southeast Missouri showing sample location (S 1/2, NW 1\4, NE 1\4, Sec. 36, T. 33 N., R. 5 E., Madison County) and inferred distribution of the planar stromatolite and burrowed carbonate mud facies of the Bonneterre Formation, along with Precambrian exposures [after Howe, 1968].

Bonneterre [Gerdemann and Myers, 1972; Thacker and Anderson, 1977; Grundmann, 1977; Larsen, 1977; Lyle, 1977; Davis, 1977; Roedder, 1977; Frank, 1981; Frank et al., 1982]. However, three paleomagnetic studies have attempted to date the age of sulfide mineralization in the MVT deposits in southeast Missouri [Beales et al., 1974; Wu and Beales, 1981; Wisniowiecki et al., 1983]. Of these three studies, only the last made extensive use of demagnetization techniques and analyzed a statistically significant number of samples. One paleomagnetic study has also been done on the underlying Lamotte sandstone [Al-Khafaji and Vincenz, 1971].

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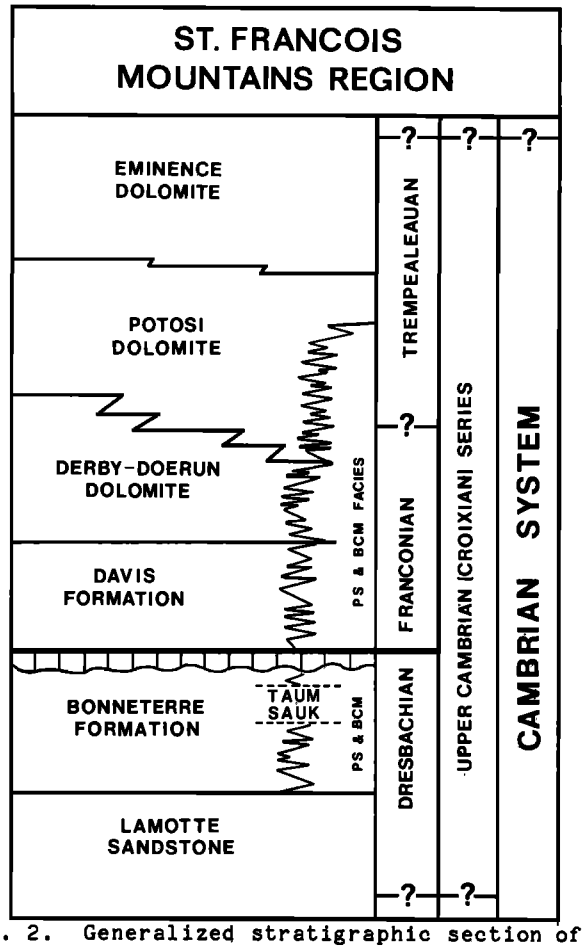


Fig. 2. Generalized stratigraphic section of the St. Francois Mountain Region, southeast Missouri, modified after Thacker and Anderson [1977]. PS and BCM refer to planar stromatolite and burrowed carbonate mud facies.

Sampling and Laboratory Techniques

The Taum Sauk at the sampling location consists of alternating silty laminated (stromatolitic) layers (0.10-0.40 m) and massive pink limestone layers (0.25-1.50 m) containing variable amounts of burrowing. The burrows, which are usually more distinct in the upper one half to two thirds of the beds, contain calcite spar commonly intergrown with iron oxides and clay minerals. The host limestone is characterized by very fine grained calcite with iron oxide-stained peloids and rare fossils.

Over 100 oriented cores were drilled and collected from 17 sites within a 3.0-m stratigraphic interval and lateral separation of less than 70 m. Several oriented rock slabs were also collected. All sites were drilled with a portable drill in the freshest possible exposures to minimize the possibility of alteration by surface weathering and were oriented with a Brunton compass. The sites represent different lithologies ranging from massive stromatolitic to well-burrowed limestone. Each site was located in a single bed of one lithology. All beds in the quarry are flat lying.

Samples were measured using a United Scientific three-axis cryogenic magnetometer. Alternating field (af) demagnetization, using a two-axis tumbling af demagnetizer, did not remove significant magnetization except in the

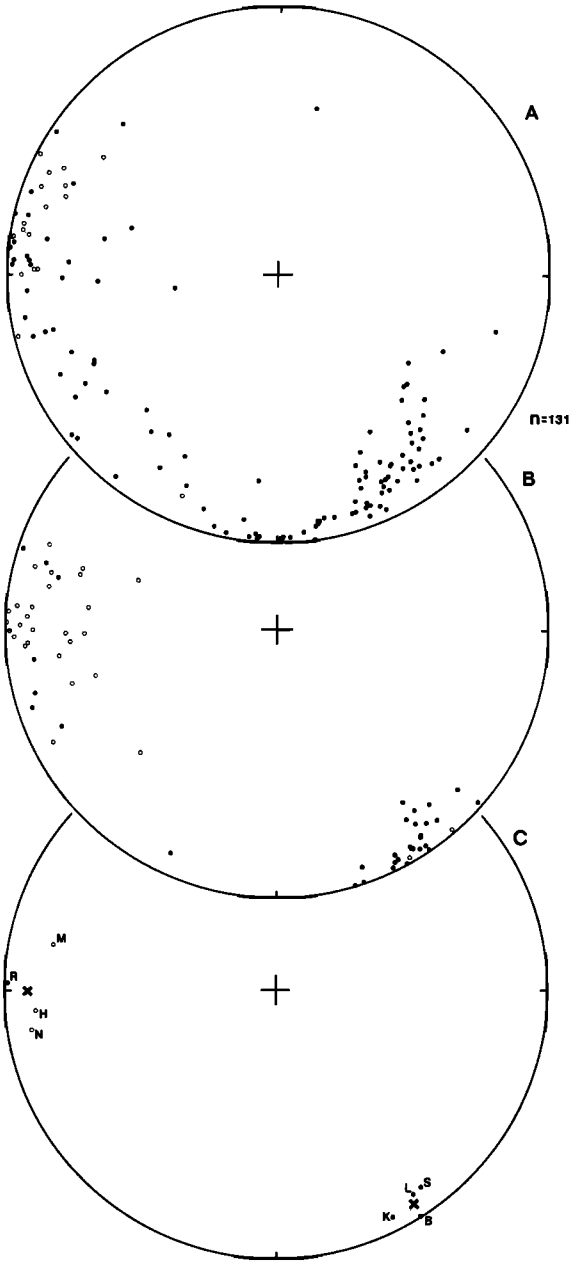


Fig. 3. Equal-area plots of (a) NRM directions of all specimens, (b) vector subtraction directions obtained through thermal demagnetization of specimens in groups one and two (notice grouping), and (c) site mean directions from groups one and two. Crosses denote group means for groups one and two (see Tables 1 and 2 for statistical information). Upper case letters refer to sampling site mean directions (H, M, N, and R are in group one; B, K, L, and S constitute group two). (Open circles have negative inclination, solid circles have positive inclination.)

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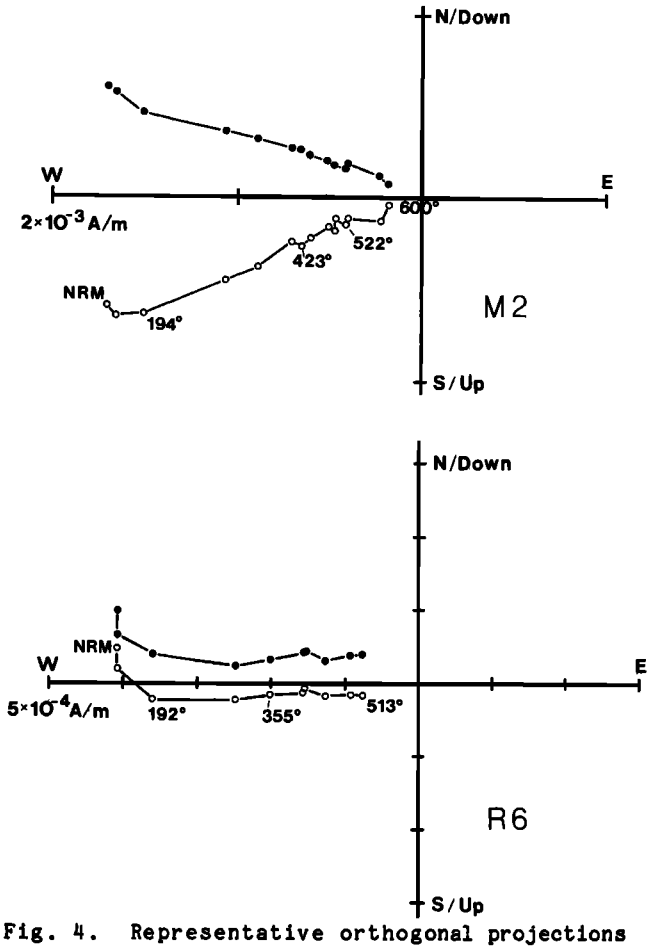


Fig. 4. Representative orthogonal projections of remanence decay during thermal demagnetization from group one. Open circles are projections on the vertical plane, and solid circles are projections onto the horizontal plane.

case where a large viscous remanent magnetization (VRM) resided in a specimen. Thermal treatment of the specimens was the most useful demagnetization technique. Stepwise thermal demagnetization was conducted until a stable end point or a stable direction of decay could be established. In most cases, over 90% of the original remanence was removed. Specimens were heated to 300°C in 100°C intervals and then in 20°-25°C intervals with a Schonstedt I thermal demagnetizer (thermal gradients were mapped within the oven and temperatures were corrected for each demagnetization step, and the residual field was less than 5 nT during cooling of specimens). During each heating interval, cores were allowed to heat for 30 min in order to equilibrate, were cooled to room temperature, and then were measured. Following treatment, results for each specimen were plotted in orthogonal projections [Zijderveld, 1967]. Individual components of magnetization were determined from the orthogonal projections for linear demagnetization trajectories defined by four or more points and then applying vector substraction to the results that defined the linear trends. Site mean directions were computed from specimen directions in each site. Component directions of magnetization were calculated giving equal weight to specimen directions. Selective destructive demagnetization (SDD) [Larson, 1981], involving mechanical removal of burrows containing iron oxides, calcite cement, and clay was also successful in some cases for removing directions of magnetization.

Further, use of coercivity spectrum analysis as described by Dunlop [1972] and compared with results obtained for other limestones [Lowrie and Heller, 1982] helped verify the magnetic mineralogy present in the Taum Sauk.

Thin section analysis with transmitted and reflected light as well as cathodoluminesence petrography was conducted to identify the important mineral phases such as calcite, clay, dolomite, and the opaque iron oxide minerals and to determine the textural relationships among the different phases. Samples were also examined using a scanning electron microscope (SEM) with energy dispersive capability.

Paleomagnetic Results

NRM directions for all specimens are shown in Figure 3a. On the basis of results from laboratory analysis the sites can be divided into three groups. Group one contains sites dominated by a westerly and shallow direction of magnetization, whereas group two consists of sites dominated by a southeasterly and shallow direction. Group three sites contain specimens

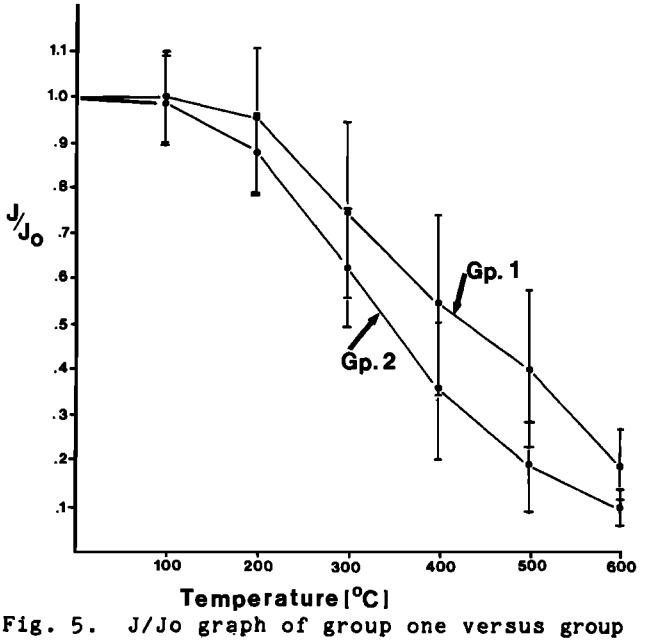


Fig. 5. J/Jo graph of group one versus group two thermal decay shows that the group one component has a higher mean blocking temperature range than the group two component. The southeasterly magnetization in group two is removed at lower mean temperatures, leaving a greater percentage of the westerly component (group 1) to dictate the magnetic signature of the rock at higher temperatures. Error bars are standard deviations for each data point (group one mean - 33 specimens; group 2 mean - 23 specimens).

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TABLE 1. Paleomagnetic Site Mean Directions

Site	Dec./Inc., deg	N/No	k	α95	Pole, deg	dp, deg	dm, deg
H	226/-12	13/13	17.0	10.4	7.2S, 177.1E	5.4	10.5
M	282/-18	8/8	22.8	11.9	3.7N, 165.0E	6.4	12.3
N	261/-10	8/8	6.2	24.2	10.4S, 179.0W	12.4	24.5
R	272/ 2	7/7	30.2	11.2	12.0N, 178.9E	5.6	11.2
B	148/ 0	10/10	65.6	6.0	41.9N, 135.7E*	3.0	6.0
K	153/ 13	3/4	77.9	14.1	39.1N, 125.7E*	7.3	14.3
L	145/ 17	9/9	34.9	8.8	33.8N, 132.0E*	4.7	9.1
S	143/ 17	5/5	150.0	6.3	32.4N, 134.4E*	3.3	6.5
C	164/ 8	7/7	56.2	8.1	45.8N, 113.4E*	4.1	8.2
D	199/ 0	5/5	12.7	22.3	48.3N, 59.9E*	11.2	22.3
E	173/ 10	3/3	19.0	29.1	46.9N, 99.4E*	14.9	19.5
F	165/ 3	7/7	29.8	11.2	48.7N, 112.4E*	5.6	11.2
I	163/ 24	3/3	4.3	68.8	37.3N, 110.6E*	39.4	73.6

Dec. and Inc. are declination and inclination, respectively; N/No is the ratio of specimens or sites used in the statistical analysis to the number of samples studied; k is the statistical parameter [Fisher, 1953] associated with the specimen or site mean; α95 is the associated radius in degrees of the cone of 95% confidence; dp and dm are semiminor and semimajor aces, respectively, of the oval of 95% confidence about the poles associated with the paleomagnetic directions.

*The pole is reversed 180° from the actual calculated pole position.

that plot on an arc between the group one and group two directions and move toward the group one direction during thermal demagnetization. Thirteen of the 17 sites collected fit the groupings above. The other four either had NRM intensities too low to yield stable demagnetization or were dominated by highly viscous components.

Group one consists of four sites (H, M, N, and R) which all show consistent thermal decay of the westerly direction of magnetization. NRM intensities for these sites range from 2×10^{-3} to 1.7×10^{-4} A/m. The linear decay expressed in orthogonal projections of specimens M2 and R6 is typical of group one (Figure 4). A J/Jo graph of average intensities for all group one specimens during thermal demagnetization shows a maximum decline of remanence intensity from 200° to 500°C (Figure 5). Figure 3b shows the directions for all specimens in sites H, M, N, and R. From these the H, M, N, and R site mean directions were computed, which are shown in Figure 3c and listed in Table 1. The group one direction has declination 270° and inclination -11° (Figure 3c) and corresponds to a pole at latitude 3.6°S, and longitude 175.6° E (Table 2).

On the basis of coercivity analysis, af, and thermal demagnetization, hematite appears to be the only magnetic mineral present in group one. No significant decay resulted from af treatment up to 0.35 T peak field. Peak fields of this magnitude exceed most coercivities of magnetite. Isothermal remanent magnetization (IRM) of core R10 fails to reach a maximum in fields up to 0.7 T (Figure 6a) consistent with the predominance of hematite over magnetite. The presence of hematite is further supported by blocking temperatures above 600°C of the IRM induced by a 0.7-T field for 30 s.

Group two consists of four sampling sites (B, K, L, and S) which show consistent thermal decay of the southeasterly direction of magnetization. NRM intensities for group two are generally stronger than those for group one and range from 5×10^{-3} to 3.6×10^{-4} A/m. NRM directions generally plot in the vicinity of declination 145° and inclination 36° (Figure 3a). Upon heating, these directions become more shallow, but their declinations remain about the same with a slight clockwise shift of 4° which may be due to the removal of a modern VRM. This decay is shown in the orthogonal projections of specimens L5 and S1 (Figure 7). The most

TABLE 2. Taum Sauk Directions

Site	Dec./Inc., deg	N/No	k	α95	Pole, deg	dp, deg	dm, deg
West	270/-11	36/36	12.3	7.1	3.6S, 175.6E	3.6	7.2
SE	147/ 10	27/28	38.0	4.6	37.3N, 133.1E*	2.3	4.6

See Table 1 footnotes.

*The pole is reversed 180° from the actual calculated pole position.

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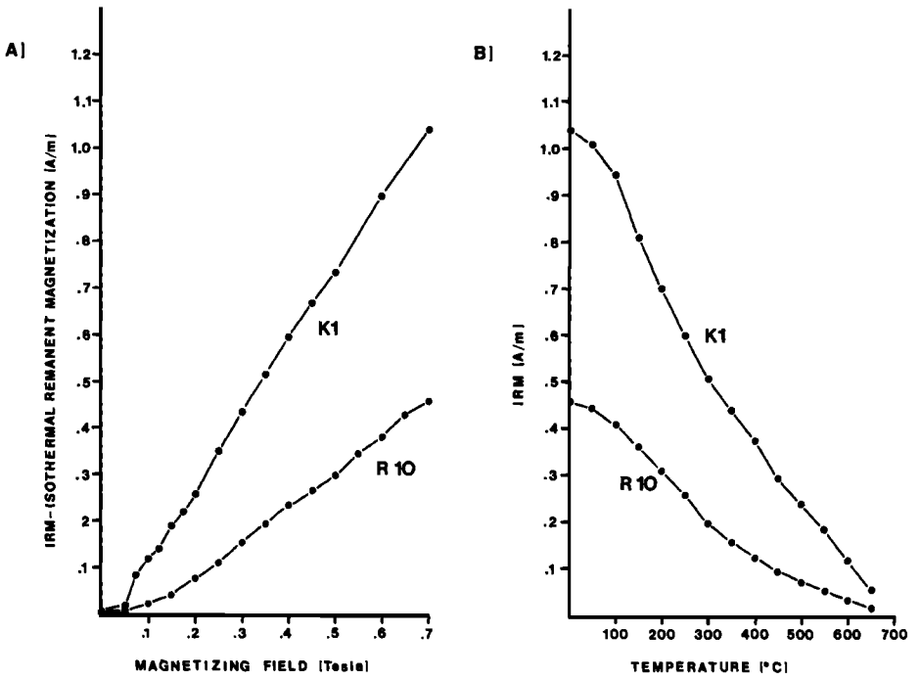


Fig. 6. IRM acquisition and demagnetization results from two specimens, K1 (group 2) and R10 (group 1).

rapid thermal decay occurs between 200° and 400 °C, and the decay is more rapid than for the group one magnetization which has a higher mean blocking temperature range (Figure 5). Vector subtraction directions from all group two specimens are shown in Figure 3b. The site mean directions obtained from the vector subtractions are shown in Figure 3c and listed in Table 1. The group mean direction calculated from specimen directions has declination 147° and inclination 10 ° (Figure 3c) corresponding to a reversed pole position at latitude 37.3°N and longitude 133.1°E (Table 2).

Results of coercivity studies and af demagnetization on these sites are nearly identical to those obtained for group one (Figure 6). Again, hematite is the sole carrier of magnetic remanence.

Petrographic analysis of specimens from this group show a large amount of diagenetic hematite intergrown with calcite in the burrows. SDD was performed on several group two specimens. By drilling out the burrow-filling material in a stepwise manner and measuring the specimen at successive intervals, it was determined that a significant percentage of the southeasterly component resides in the hematite within the burrows. This is illustrated by the decay of specimen B3 in Figure 8.

Group three is composed of sampling sites C, D, E, F, and I which contain specimens whose NRM inclinations are less than 40° and whose declinations vary from 140° to 260°. The majority of the specimens in group three have NRM directions similar to those from group two, plotting in the southeast. NRM intensities from group three range from 3×10^{-4} to 1×10^{-4} A/m.

The most important fact about the specimens composing group three is their unique behavior during thermal demagnetization. Upon stepwise thermal demagnetization, specimens from these sites show a shift in declination from southeast to west with end points near the directions of group one. In most cases a specimen direction will first undergo a decrease in inclination and then move through the described declination shift. In all cases the declination shift occurs along an equatorial path with inclination remaining shallowly up or down (Figure 9). Though group three sites become more westerly upon thermal demagnetization, they yield directions from vector subtraction which plot closer to those of group two. Neither a direction of magnetization nor a pole position was determined for this group. However, as discussed below, vector subtraction and stable end point analysis of these sites was used to help confirm the southeasterly (group two) and westerly (group one) directions, respectively. Results of coercivity analysis and blocking temperatures are consistent with the results from groups one and two that all magnetization resides predominantly in hematite.

Lithologic and Petrographic Results

The separation of sampling sites into three groups is also supported by the lithologic and petrographic evidence. Group one (sites H, M, N, and R), which is dominated by the western direction of magnetization, is primarily composed of peloidal wackestones and packstones. Allochems are rare, and the peloids are stained with hematite (Figure 10a). Group two (sites B, K, L, and S), which is dominated by the southeastern direction, shows the same matrix characteristics as group one but contains abundant burrowing (Figure 10b). Hematite is common within the calcite that fills the burrows. Finally, group three (sites C, D, E,

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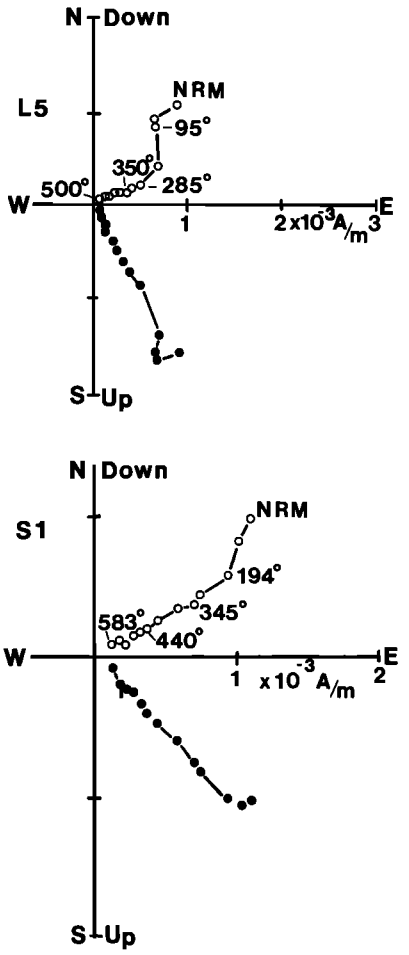


Fig. 7. Orthogonal projections of thermal decay for specimens L5 and S1 from group two. Diagrams show removal of a southeasterly and downward dipping direction of magnetization. Symbols are the same as in Figure 4.

F, and I) contains the same host matrix characteristics as groups one and two but contains a moderate amount of burrowing. Hematite is the only magnetic mineral which has been identified petrographically within any of the samples.

Numerous diagenetic features occur within the Taum Sauk Limestone, including clay, calcite, hematite, dolomite, dedolomite, and stylolites. Illite, iron oxides, and calcite are texturally interrelated within the burrows (Figure 11a). Authigenic illite occurs concentrated near the edges of burrows and also within the calcite cement (Figure 11b). Illite is also disseminated within the matrix of the limestone. Iron oxides occur associated with the illite, occurring as plates arranged in clusters (Figure 11c) and also as fine-grained pigment that stains the illite (Figure 11d). This pigment could not be resolved with the SEM at 50,000X. The fabric in which the calcite, iron oxides, and illite are associated (Figure 11d) is similar to certain other known intergrowth fabrics but might also be a desiccation fabric. In any case, the three phases were probably precipitated before the development of the dolomite that occurs in the centers of the burrows.

Two types of pore-filling calcite crystals occur: generally rare bladed crystals and more common equant crystals. Cathodoluminescence petrographic studies have revealed the presence of distinctly luminescent zones within the calcite (see also Frank et al. [1982]). Iron oxides are also intergrown with the calcite cement along growth bands (Figure 11e), which suggests coprecipitation of the two phases.

Dolomite, which is rare in those samples examined, occurs in the centers of some burrows, indicating a postcalcite cement origin (Figure 11f). In highly burrowed intervals the dolomite rhombs have been selectively dedolomitized and contain intergrown calcite and iron oxides [Frank, 1981]. It was not possible to collect cores from the highly burrowed zones because they are too friable. For this reason, dedolomite was not observed in the samples used for paleomagnetic analysis.

Stylolites crosscut most other diagenetic features and probably represent one of the last stages of diagenesis. Some features, such as opaque hematite grains which occur in the matrix, could not be placed in the diagenetic sequence. The cubic shape of most opaque grains suggest that they are probably pseudomorphs after pyrite.

In summary, the textural relationships between the diagenetic features suggest the following sequence of events. Illite, iron oxides, and calcite were precipitated within the burrows. Although some of the iron oxide apparently developed during initial burrow filling, it was also precipitated in several stages during formation of the pore fill calcite. The next event was dolomitization, followed by dedolomitization and stylolitization.

Frank [1981] reported both hematite and goethite in dedolomitized crystals in the Taum Sauk. We found little petrographic evidence for goethite in samples that we examined. Color and morphology suggest that the dominant iron

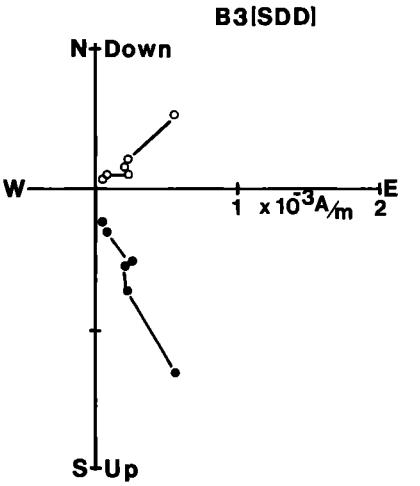


Fig. 8. Orthogonal projection of SDD on specimen B3 (group 2) shows removal of the same component of magnetization as yielded by thermal demagnetization. Steps represent removal of burrow-filling material at core surface and then progressive removal of deeper burrow-filling material. Symbols are the same as in Figure 4.

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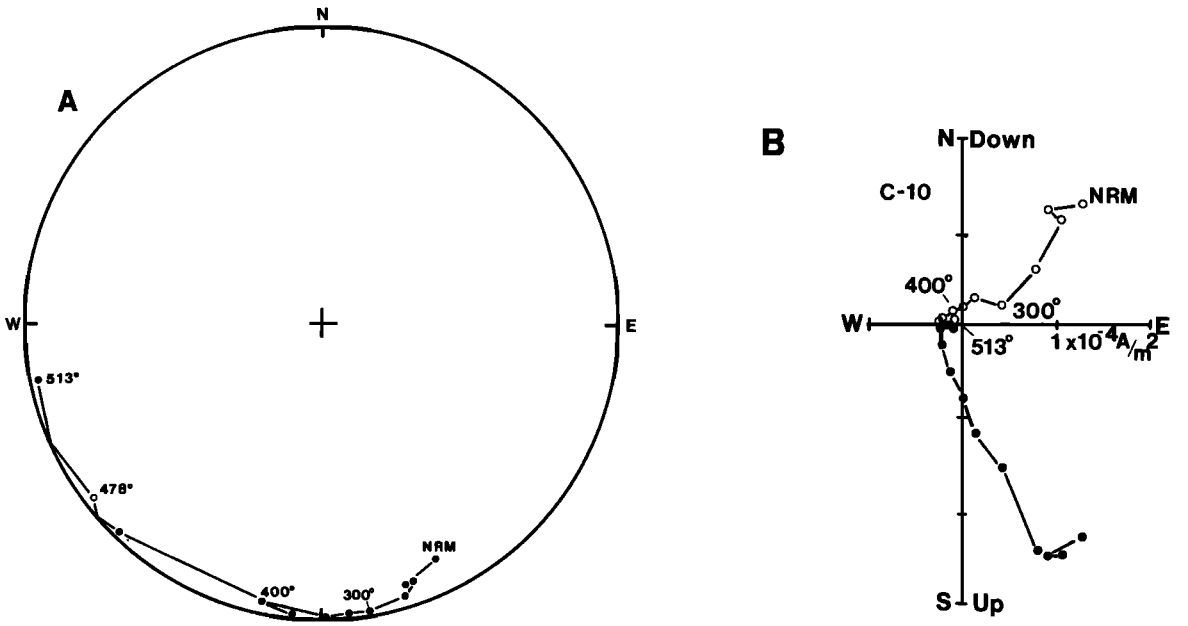


Fig. 9. (a) An equal-area projection of specimen C-10 during thermal demagnetization (b) The corresponding orthogonal projection. C-10 starts with a southeasterly declination but becomes westerly with increased temperature. Inclination remains shallow throughout. The main component of magnetization removed during demagnetization has a southeasterly declination and shallow positive inclination (solid circles indicate positive inclination; open circles represent negative inclination). The vector being removed is most likely a combination of a dominant southeasterly direction and a weak westerly direction. As the maximum blocking temperature of the southeasterly component is reached (about 500°C) another direction of decay is shown in Figure 9b. This is the removal of the westerly direction which is still unblocking at 513°C.

oxide is hematite. As previously presented, the paleomagnetic and rock magnetic results support this conclusion.

Discussion

The Taum Sauk Limestone contains two directions of magnetization residing in hematite. As discussed below, the group one (westerly) direction of magnetization is probably Late Cambrian, whereas the group two magnetization (southeasterly) corresponds well to the late Paleozoic APWP for North America but might also fit the Cambrian APWP suggested by Watts et al. [1980]. Both directions appear to be CRM's which resulted from the precipitation of diagenetic hematite. Paleotemperature estimates from the Bonneterre Formation yield maximum temperatures near 120°C [Roedder, 1977]. On the basis of a relationship between relaxation time and temperature [Pullaiah et al., 1975], 120°C is too low for either component to be a thermal remagnetization of preexisting hematite.

The pole position obtained from the group one sites in the Taum Sauk at latitude 3.6°S, longitude 175.6°E (Table 2) is similar to some poles interpreted to be Cambrian (Table 3). This Taum Sauk pole, however, does not correspond well to other poles interpreted to be Cambrian nor does it correspond to the Cambrian APWP loop for North America proposed by Watts et al. [1980] (Figure 12a). Such discrepancies are not surprising when one considers that paleomagnetic poles from North American Cambrian rocks plot over almost one half of a hemisphere and few are age constrained by any type of field test.

Consider first the two original paleomagnetic poles from the conformably underlying Lamotte Formation [Al-Khafaji and Vincenz, 1971]. Pole positions at 160°W, 38°S and 155°E, 11.5°S were obtained through af demagnetization, yet the magnetization in the Lamotte is apparently carried by hematite [Al-Khafaji and Vincenz, 1971]. The af demagnetization procedure is often a poor way to resolve magnetic directions residing in hematite, and as mentioned by Gillett [1982], the direction of magnetization observed in the Lamotte study may be the vector sum of unresolved components. Since the Al-Khafaji and Vincenz study, two poles have been cited from the original Lamotte data using thermal cleaning to locate a pole at 168.0°E, 0.5°N and af cleaning to locate a pole at 177.4°E, 8.4°S [Van der Voo et al., 1976]. These poles bear a much closer relation to the group one pole position of the Taum Sauk (Figure 12a). Two other studies of similarly aged units yield paleomagnetic poles close to the group one Taum Sauk pole position. The Wilberns Formation (Late Cambrian), Texas yields a pole at 158.8°E, 6.4°N [Van der Voo et al., 1976] and the Wichita Granites (Cambrian) have a pole at 164°E, 4°N [Spall, 1970] (Table 3). The Taum Sauk pole also falls close to the Cambrian section of Irving's [1979] APWP.

Considering the complexity of the

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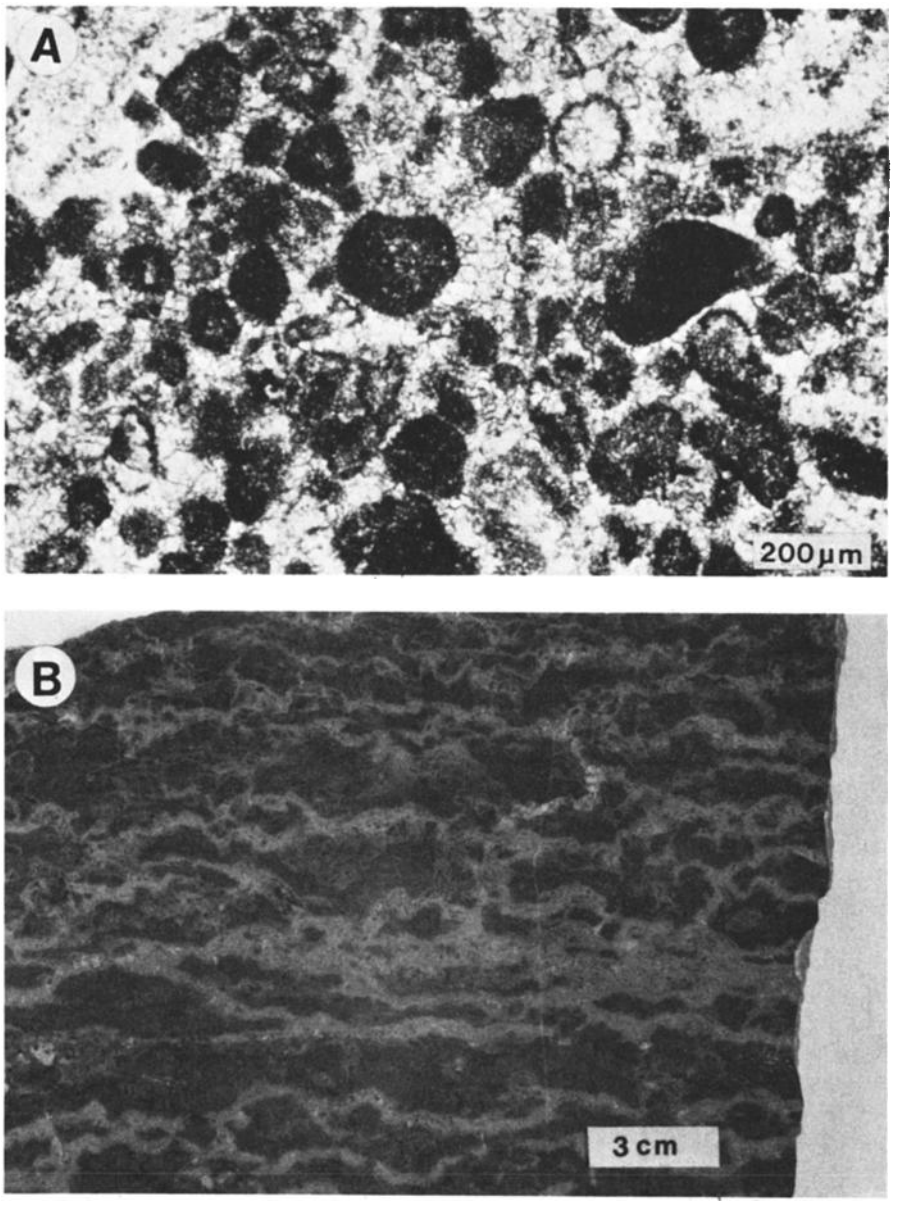


Fig. 10. Lithologic and petrographic features in the Taum Sauk Limestone. (a) H site specimen (group 1) shows limestone matrix of fine calcite, rare allochems, and peloids stained with hematite. (b) Slab of pink limestone that is moderately to extensively burrowed. Burrows (light colored stringers) contain hematite, clay minerals, and calcite spar.

magnetization in the Taum Sauk, it is important to take into account the possible effects of subsequent remanences with overlapping blocking temperatures on the component directions and corresponding pole positions. For example, if the westerly component was partially superimposed by a southeasterly component, the group one pole position would be shifted away from many of the Cambrian poles listed in Table 3.

Assuming from the foregoing argument that acquisition of the group one direction in the Taum Sauk occurred in the Late Cambrian, several origins of magnetization are possible. One likely origin of the magnetization is the same as for the "Scaglia" limestones of Italy [Channell et al., 1982], where it was postulated that the hematite which carries the remanence formed from a goethite precursor during early diagenesis. The goethite precipitated directly from seawater but dehydrated to hematite during early diagenesis because it is unstable under most geologic conditions [Berner, 1969]. This mode of acquisition for the Taum Sauk fits the proposed Late Cambrian magnetization. A second possible origin for the group one remanence is a DRM or a postdepositional remanent magnetization (pDRM). Given that the depositional environment was very close to an exposed igneous source, it is possible that fine-grained detrital hematite was incorporated into the lime muds during deposition. Although no detrital specular hematite was discovered petrographically, hematite as well as magnetite are present in the underlying igneous rocks

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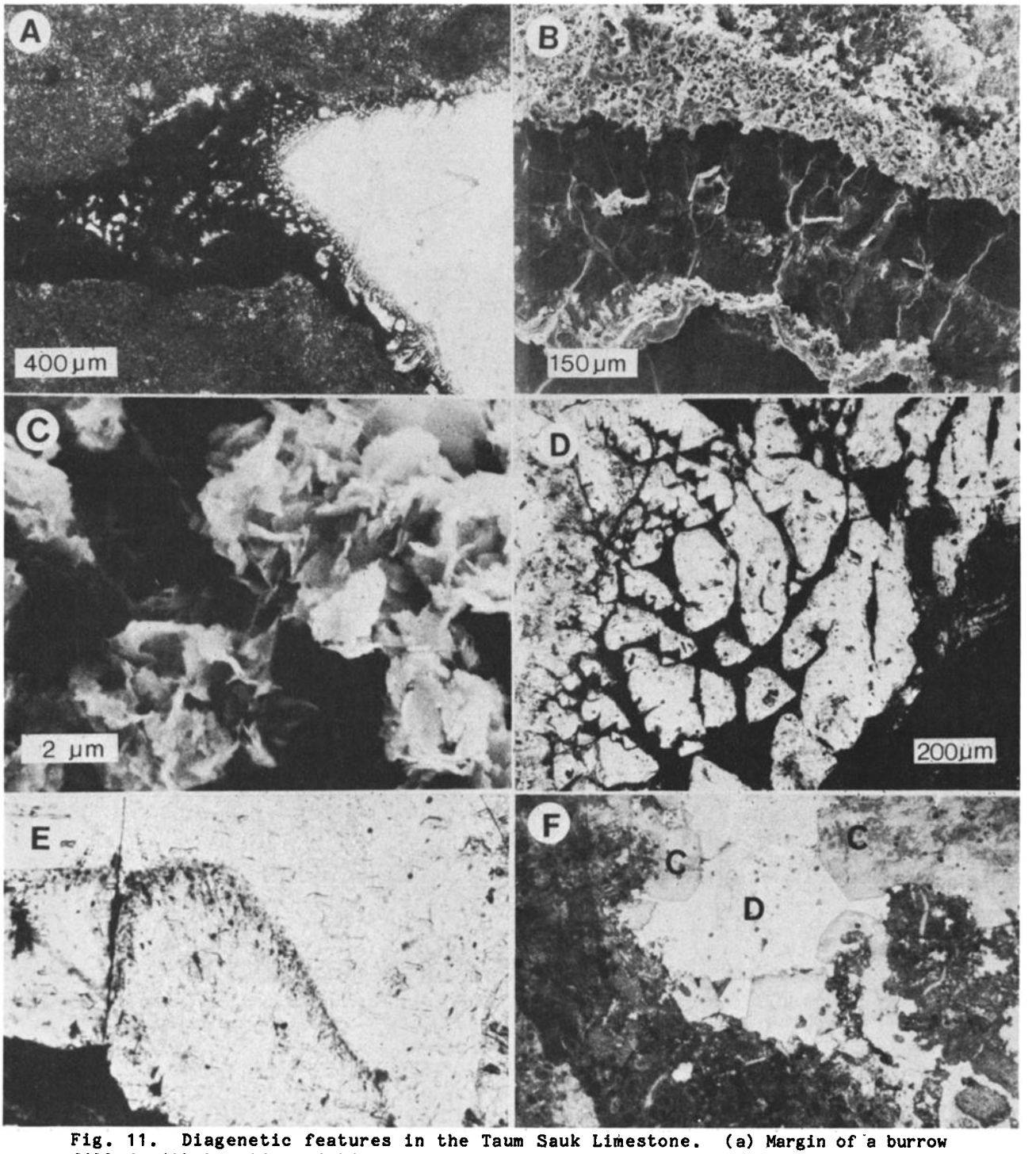


Fig. 11. Diagenetic features in the Taum Sauk Limestone. (a) Margin of a burrow filled with hematite, illite, and calcite. Dark material is hematite-stained illite (plane polarized light). (b) SEM photomicrograph showing illite and calcite at the margin of a burrow. (c) SEM photomicrograph of hematite clusters associated with illite, which surrounds the hematite in the background. Sample was etched for 15 s in 10% HCl. (d) Calcite (clear) associated with hematite/illite. Dark material is hematite and illite (plane polarized light). (e) Hematite within calcite crystals from the edge of a burrow (plane polarized light). (f) Burrow with calcite. C (dark crystals), calcite, stained with Alizarin red-S, near the margin of a burrow; D, dolomite in the center part of the burrow.

[Kisvarsanyi, 1977]. Because of the amount of burrowing present in the unit, considerable bioturbation must have occurred after deposition. This reworking should have kept the water content above the critical level to prevent locking of a DRM [Verosub, 1977]. Thus a pDRM acquired after cessation of soft sediment reworking is a possibility. The magnetization could also be carried by hematite (martite) that replaced detrital magnetite during early diagenesis. An additional possibility is that the hematite which pseudomorphically replaced

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TABLE 3. Paleomagnetic Poles From Late Cambrian Formations

Figure 11	Formation	Pole	Age of Acquisition	Reference
L3	Lamotte	1°N, 168°E	C	Van der Voo et al. [1976]
L4	Lamotte	8°S, 177°E	C	Van der Voo et al. [1976]
B3	Bonneterre	38°N, 144°E	Penn.-Perm.	Wu and Beales [1981]
B4	Bonneterre	41°N, 126°E	Penn.	Wisniowiecki et al. [1983]
W	Wilberns	6.4°N, 158°E	C	Van der Voo et al. [1976]
WG	Wichita Granites	4°N, 164°E	C	Spall [1970]
RH	Riley (Hickory)	33°N, 140°E	C	Watts et al. [1980]
RL	Riley (Lion Mountain)	27°N, 146°E	C	Watts et al. [1980]
TS1	Bonneterre	4°S, 176°E	C	this study
TS2	Bonneterre	37°N, 133°E	Penn?	this study

pyrite may carry the magnetization. If this is the case, then the process probably occurred during early diagenesis.

The southeasterly component of magnetization in the Taum Sauk represented by group two sites yields a pole at latitude 37.3°N, longitude 133.1°E after reversal. This pole correlates well with the Carboniferous portion of the APWP for North America (Figure 12b) but also correlates to the Cambrian loop proposed by Watts et al. [1980] (Figure 12a). Initially, either a Cambrian or late Paleozoic age of acquisition seems possible. Although the existence of the loop has been questioned [Gillet, 1982; Tarling, 1983], several arguments suggest that a Cambrian age is unlikely. First, e en if the loop is correct, the secondary Taum Sauk paleomagnetic pole corresponds to the wrong section of the Cambrian loop based on the stratigraphic age of the Taum Sauk (upper Dresbachian) and the paleomagnetic poles used to construct the loop. On the basis of the loop the earliest and latest Cambrian poles should plot near 10°N, 160°E, with the middle Cambrian poles near 50°N, 100°E. Poles from the Riley Formation (Hickory and Lion Mountain members), Texas, were used to construct the loop (Figure 12a). The Hickory member is lower Dresbachian whereas the Lion Mountain member is upper Dresbachian. The poles from these units plot to the south and east of the Taum Sauk pole in Figure 12a. This suggests that the second Taum Sauk pole does not correspond to the Cambrian loop. Second, two other paleomagnetic poles from the Bonneterre have been determined to represent late Paleozoic (Pennsylvanian-Permian) remagnetization and plot close to the group two pole obtained for the Taum Sauk (Table 3). The surface-subsurface study of the Bonneterre in the Viburnum Trend [Wisniowiecke et al., 1983] documented a pole at 126°E, 41°N, with the magnetization probably residing in diagenetic magnetite spheroids (Table 3). The Pennsylvanian age of acquisition for this pole is constrained by a breccia test. Magnetization was demonstrated to be postbrecciation, which is thought to have occurred during the Ordovician. Wu and Beales [1981] also reported a Pennsylvanian-Permian paleomagnetic pole believed to be associated with lead mineralization processes (Table 3). These paleomagnetic poles (B3 and B4) are closer to the group two Taum Sauk pole than any others cited (Figure 12b), and although reflecting different processes of remagnetization, their similar paleomagnetic poles suggest a similar remagnetization time as for the Taum Sauk.

The group two component of magnetization in the Taum Sauk is most likely late Paleozoic. As with group one, the magnetization resides in diagenetic hematite. This hematite is most visible in burrows in the limestone. Here hematite is intergrown with calcite suggesting coprecipitation. Thus CRM is the postulated mode of remanence acquisition. On the basis of the petrographic relations, any dolomitization and dedolomitization occurred after coprecipitation of calcite and hematite. The paleomagnetic data for the calcite and hematite constrain the age of dolomitization and dedolomitization in the samples to Carboniferous or younger.

Group three samples are both lithologically and paleomagnetically intermediate in character between groups one and two. Group three sites have more burrowing than group one sites yet less than group two sites. Magnetically, group three specimens contain both components of magnetization. Vector subtraction directions from specimens in group three are generally similar to those from group two, yet their end points from high-temperature cleaning are similar in position to group one results. The key to interpreting group three's magnetization is to be able to distinguish between the westerly and southeasterly components of magnetization which are represented in groups one and two, respectively. The NRM intensities are stronger for the southeasterly component than for the westerly component, which is caused by a greater abundance of hematite in group two specimens. As a result, the southeasterly component will usually be dominant when both magnetizations reside in the rock. The group two mean blocking temperature spectrum, however, is skewed toward lower temperatures than group one (Figure 5). At high demagnetization temperatures, the westerly direction is more dominant as the southeasterly direction is removed at lower temperatures (Figure 9).

Sedimentologic Effects on Remagnetization

The burrows in the Taum Sauk Limestone played an important role in the remagnetization of the unit. Apparently, the burrows were the main avenues for the fluids that precipitated the

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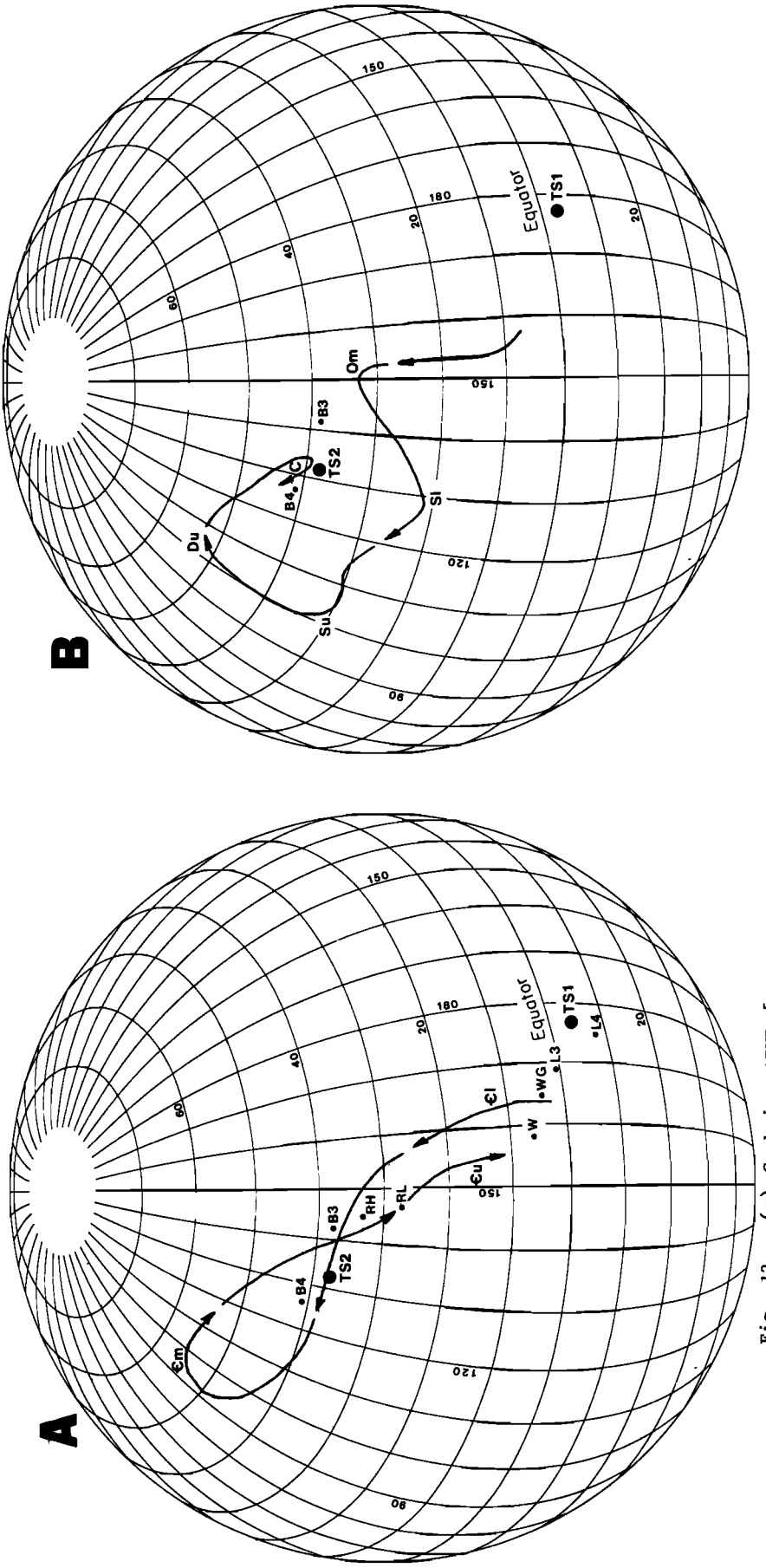


Fig. 12. (a) Cambrian APWP [Watts et al., 1980]. Poles TS1, L3, L4, W, and WG plot in a likely location for Cambrian poles. TS2 is the pole from the secondary Taum Sauk component (Table 3). B3 and B4 are Penn-Permian poles from the Bonneterre (Wu and Beales [1981] and Wiśniowiecki et al. [1983], respectively). RH (Riley, Hickory member) and RL (Riley, Lion Mountain member) were used to construct Watts' Cambrian loop. They are lower and upper Dresbachian, respectively. TS2, B3, and B4 all fit the wrong portion of Watts' loop for them to be upper Dresbachian poles. (b) Paleozoic APWP [Van der Voo, 1981]. TS2, B3, and B4 all correspond with the Carboniferous portion of the Paleozoic APWP by Van der Voo [1981] shown here.

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diagenetic hematite. In beds dominated by burrows (group two sites) the fluids precipitated diagenetic hematite that totally remagnetized the rock. In beds without burrows there were no avenues for fluid percolation, and these sites (group one sites) were not remagnetized in the late Paleozoic, retaining a Cambrian direction of magnetization. Beds with an intermediate amount of burrowing (group three sites) contain both the Cambrian and Carboniferous components of magnetization.

The fluids that precipitated diagenetic hematite and caused the remagnetization of the Taum Sauk apparently had a different chemistry than other fluids which affected the Bonneterre Formation, which included those that precipitated the diagenetic magnetite [Wisniowiecki et al., 1983] and the fluids that caused the sulfide mineralization. In addition to differences in fluid chemistry the relative chemistries of the host rocks may also have been important. For example, most sites sampled by Wisniowiecki et al. [1983] were in the reef complex, which contains the main bodies of sulfide ore. The lagoonal or backreef rock represented by the Taum Sauk is quite different from the reef rock. The diagenetic history of the reef rock [Lyle, 1977] is also different than for the Taum Sauk Limestone.

The possibility of late Paleozoic remagnetization of early Paleozoic rocks has long been a controversial topic. To study this problem, Gillett [1982] sampled across the Crepicephalus-Aphelaspis trilobite boundary in the Nolichucky Formation (east Tennessee) and compared the results to those obtained in other studies on rocks at or near this time line across North America. One of the studies cited was a study of the Bonneterre [Beales et al., 1974] because of evidence of a Crepicephalus fauna reported in the unit [Lochman-Balk, 1971]. Of the five studies cited by Gillett, one shows conclusive evidence of remagnetization [Gillett and Van Alstine, 1982], three suggest remagnetization or tectonic rotation [Gillett et al., 1982; Watts et al., 1980; Elston and Bressler, 1977], and the fifth is the Bonneterre [Beales et al., 1974]. Gillett suggested that the study by Beales et al. [1974] is inconclusive and calls for a mroe thorough investigation of the Bonneterre. On the basis of this study of the Taum Sauk and the study of the Bonneterre by Wisniowiecki et al. [1983], remagnetization of the Bonneterre is apparent. Further, the Pennsylvanian is the likely time of remagnetization.

This Taum Sauk study incorporates extensive sampling of a single unit at one location, and it shows the effect that diagenetic and sedimentologic processes can have on the magnetization of sedimentary rocks. Six sites (C, E, H, M, N, and P) with a maximum lateral separation of 35 m, are from one 40-cm interval. Three of these sites exhibited group one remanence with the westerly directions. Two sites represented group three intermediate directions. One site had NRM intensities too weak to yield results from magnetic cleaning. Thus incomplete sampling of the different limestone types in this quarry could have resulted in an incorrect or incomplete interpretation of the magnetic history of these rocks. Future paleomagnetic work on units such as the Taum Sauk should emphasize linking the magnetic and petrographic characteristics.

Conclusions

A thorough sampling of one quarry in the Taum Sauk Limestone was conducted for a paleomagnetic investigation of the unit and has lead to the following conclusions:

1. On the basis of microscopic observations, thermal demagnetization, and coercivity spectrum analysis, all magnetization resides in hematite.
2. Two stable remanences with overlapping blocking temperatures were isolated in the unit. On the basis of thermal history and petrographic analysis, both components are CRM's and resulted from diagenetic processes.
3. One component has declination 270°, inclination -11° corresponding to a Cambrian pole position at 175.6°E, 3.6°S. The hematite which carries this remanence might be the result of dehydration of a goethite precursor.
4. The second stable component has declination 147°, inclination 10° and corresponds to a reversed pole at 133.1°E, 37.3°N, which suggests a Pennsylvanian age of magnetization. The hematite carrying the secondary component appears intergrown with calcite and constrains the age of dolomitization in the burrows to Pennsylvanian or younger.
5. Finally, the presence of two stable directions of magnetization in one location with neither being pervasive throughout the quarry shows that diagenetic and sedimentologic processes can have a pronounced effect on the magnetization of sedimentary units. Only through extensive sampling of all limestone types in the quarry were we able to unravel the magnetization history. This study also illustrates the need to link the magnetic and petrographic characteristics of sedimentary rock units.

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